Document 1 -- Proposal for MBC-3

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I. COLLABORATIVE FMCW RADAR SWARM WITH ONBOARD AI FOR AERIAL SURVEILLANCE

A. Problem Statement

Contemporary aerial surveillance requires persistent, wide-area detection of low-observable targets at operationally sustainable cost. Single-platform airborne radar is vulnerable to attrition and cannot provide distributed coverage. Aran Technologies addresses this requirement through a five-drone swarm configured as a distributed FMCW radar -- a micro-AWACS architecture -- that distributes aperture across platforms, provides mission redundancy, and enables autonomous track management independent of continuous ground-station control.

B. Proposed Solution

Aran Technologies proposes a five-hexacopter swarm, each platform carrying six 24 GHz FMCW radar panels (TI AWR1843, 60deg H-FOV per panel) providing full 360deg coverage per drone. A three-layer onboard AI pipeline autonomously processes raw radar data through to tactical decision output.

Layer 1 -- Signal Processing: The AWR1843 onboard DSP performs range-Doppler FFT and CFAR detection within 10 ms per scan cycle, producing candidate detections with range, azimuth, radial velocity, and SNR.

Layer 2 -- Target Classification: A Random Forest classifier executing on each drone's Jetson compute module classifies candidates as confirmed targets or clutter within 50 ms, using SNR, velocity, range-rate, and estimated RCS. Only confirmed detections propagate to Layer 3.

Layer 3 -- LLM Tactical Engine: Llama 3.2 3B (leader drone) and Gemma 2B (soldier drones) receive structured JSON situation reports and generate tactical commands -- track reassignment, formation reallocation, sector reorientation, and threat alerts -- operating within edge-compute budget and triggering exclusively on Layer 2 confirmations.

C. Swarm Architecture and Resilience

Command hierarchy: Ground Station > Leader Drone > Soldier autonomous mode. On leader heartbeat loss exceeding 2 seconds, the highest-battery soldier self-elects as leader via bully election protocol, assuming radar fusion and ASP publication within 2 seconds. On soldier loss, the leader LLM redistributes orphaned track IDs to the nearest active drone. Full Air Situation Picture continuity is sustained with three or more drones operational, satisfying the MBC-3 graceful degradation requirement.

A Flask-based Ground Control Station displays a consolidated real-time ASP at 2.5 Hz -- track table, polar radar display, leader identity, decision log, and timestamped JSON recording -- on a single browser screen with no external dependency.

D. Platform Specification

Each hexacopter carries: TI AWR1843BOOST radar panels x6, indigenous STM32 flight controller (custom PCB), Doodle Labs AES-128 mesh radio, and Jetson edge compute (AGX Orin 64 GB on leader; Orin NX 16 GB on soldiers). AUW <= 4.3 kg. Endurance: ~32 min (6S, 10,000 mAh). Auto RTH on data-link loss or low battery. GNSS-denied resilience via EKF2 fusing optical flow, IMU, and barometer.

E. Indigenisation

The complete intelligence layer -- CFAR software, Random Forest classifier, LLM tactical engine, ROS2 swarm coordination stack, GCS dashboard, custom antenna PCB design, and hexacopter airframe fabrication -- is 100% indigenously developed. Weighted indigenous content: >= 55% by mission-criticality, satisfying MBC-3 S.2.25.

F. Phase I Deliverable

Phase I testing complete: five-drone SITL simulation verified (ASP generation, LLM track reallocation, graceful degradation), broadband and drone-to-ground comms tested, pre-recorded 4-min ISR demo video (1920x1080) ready -- live demonstration available at New Delhi, 13-24 July 2026 (github.com/lekkalaharsha/aran-mbc3-swarm).